

Scale Using Strain Gauges and Wheatstone Bridge

Measurements Lab

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Introduction:

The objective of this project was to make a scale. The scale would be tested with a series of known mass weights and must print out the weight in grams of these masses. The scale must have a button to tare and one to switch between grams and ounces. The goal of this project was to utilize everything learned this semester with Arduino, circuitry, coding, and analyzing measurement data according to percent error and standard deviation. This was to understand how measurement devices were made and the natural error they had.

Design Process:

The group started with fully understanding the project and goals. At the start of the project the group was given two options: make a strain gauge or buying a load cell. The pros and cons were weighed and the group decided the experienced gained from soldering a strain gauge and designing a Wheatstone bridge outweighed the ease that buying a load cell would have given. After this critical decision was made, the partners designed their strain gauge, Wheatstone bridge, and supports for the scale. They decided for the strain gauge to have two strain gauges, one on the top of the beam, and one on the bottom of the beam to create a greater difference in voltage to be amplified. Since the strain gauge on the top would be in tension and the one on the bottom in compression, that would cause more change in resistance that would be amplified and cause the scale to have greater precision. In accordance with this design, the Wheatstone bridge was planned to have both strain gauges on the same side in series and a resistor and potentiometer on the other side.

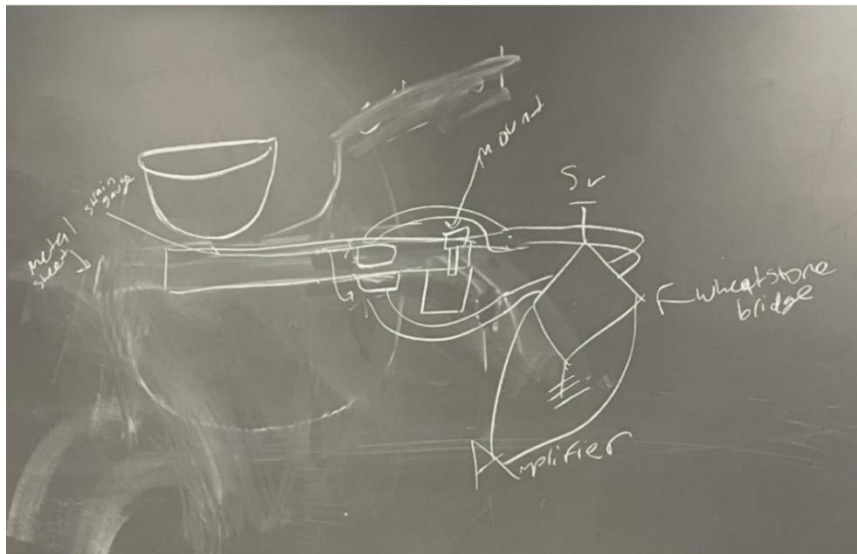


Figure 1: Initial Sketch

In testing, there were many issues in getting proper results, or any that made sense. Results were extremely variable and inconsistent. This could have been from failure in wiring or soldering (unlikely), mistakes in coding (extremely likely due to lack of understanding of given functions in HX711 library), or insufficient connections between the Wheatstone bridge and the

amplifier (as shown by messing with the wires and results dramatically changing). This last problem could only be fixed by soldering the Wheatstone bridge directly to the amplifier, a task which is very difficult and risky, or switching to a load cell and abandoning all the work put into the strain gauge and Wheatstone bridge. The group came to a crossroads, needing a quick decision on which problem to fix. They decided to split up, one taking time to fix the code, and the other preparing a load cell. The code was fixed and when testing the original design, it worked perfectly, getting consistent results. Though, the next time it was tested, the scale would not show any change in weight. The group could not figure out why so they decided to pivot. They got out the prepared load cell and set it up, using the new code. It worked and showed extremely consistent results.

System Function:

The final system consists of a load cell, a Wheatstone bridge, an HX711 amplifier and a microcontroller (Arduino). The load cell, responsible for measuring the weight, is a system of 4 strain gauges and Wheatstone bridge. A strain gauge deforms due to applied weight. As the strain gauges within the load cell deform, their electrical resistance changes. This change in electrical resistance is measured by the built-in Wheatstone bridge. A Wheatstone bridge is an electrical circuit that consists of four resistors in a diamond shape. Two of the resistors are known, while the other two are strain gauges. Each of the two branches is thus a voltage divider. Since one branch has constant and known resistor values, while the other is connected to the strain gauges, the output voltage is equivalent to 0 if the resistance in the two branches are equivalent. However, if the resistance increases in the strain gauge branch, then there will be a voltage

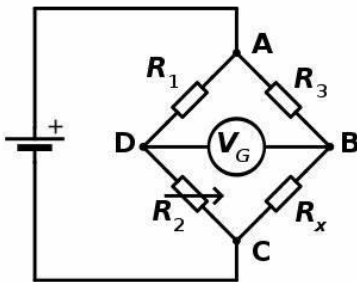


Figure 3: Wheatstone Bridge

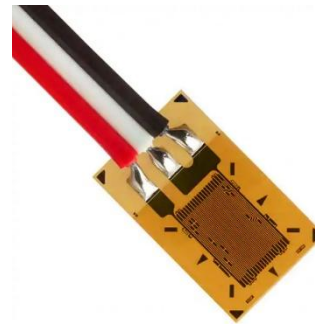


Figure 2: Strain Gauge

difference between the branches of the Wheatstone bridge.

This voltage difference is small and precise, as a result measuring weight only by using a load cell is difficult. In order to capture the voltage difference, a component called an Amplifier is

used. The team used an HX711 amplifier which is a 24 Bit Analog to Digital converter. This means that the HX711 has a resolution of 24 bits and can read up to 16777216 raw bit readings, making it ideal for precise readings. The measured values from the Wheatstone Bridge in the load cell are analog, since real-world readings are continuous, and thus, neither on or off(1 or 0). However, computers run on digital input (1 or 0), thus using HX711 provides a change in voltage reading that can be input to a computer. The HX711 provides voltage and ground connections to the load cell through E+ /E-, and the output of the HX711 is contained from pins A+/A-. The output is now digital, and the difference in voltage is digital.



Figure 4 HX711 Amplifier

To use the digital bit readings, the team connected the Arduino(microcontroller) to a computer to record the bit readings. A program which read and converted the bits to common weight units such as grams(g) and ounces(oz) was written. To record accurate readings, the team had to calibrate the scale. To do this, a known weight was measured, and the scale factor or ratio of bits to units of measurement was determined. Using the Serial.print() function within the Arduino IDE allowed for weights to be displayed.

The team was tasked with making a scale that can tare and convert between units. This was achieved using one button for the tare, and one button for g to oz conversion. Each button was placed into the breadboard and connected to ground and power. Each button has four prongs, which form two branches. Components are connected in series in each of these two branches. One branch contained a ground connection, while the other branch contained a power connection and a 100 μ F ceramic capacitor. This capacitor was then connected to a digital pin on the Arduino. The capacitor was used to prevent signals from the button being read twice and to prevent a delay in readings. The button's potential to be read twice occurred due to the mechanical error in the button. As the button is pressed it may vibrate. This vibration leads to additional (0 and 1) readings being sent to the Arduino. A capacitor stores the current resulting

from opening the switch (pressing the button), such that vibrations are ignored due to the delay in current.

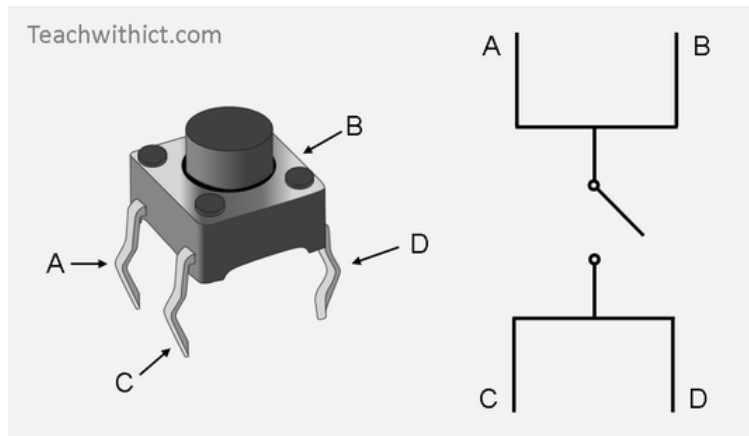


Figure 5 Button & Switch Schematic

Resistors are necessary to ensure that there is a constant signal to the Arduino and that the pins aren't "floating" (reading values that are not 0 or 1). In the team's physical circuit resistors were replaced by the Arduino's built in "Pullup resistors" which solve the issue of floating through a function within the program.

The description of the function of the system can also be explained using a Measurement Chain.

Measurement Chain:

Figure 6: Measurement Chain

Data and Analysis:

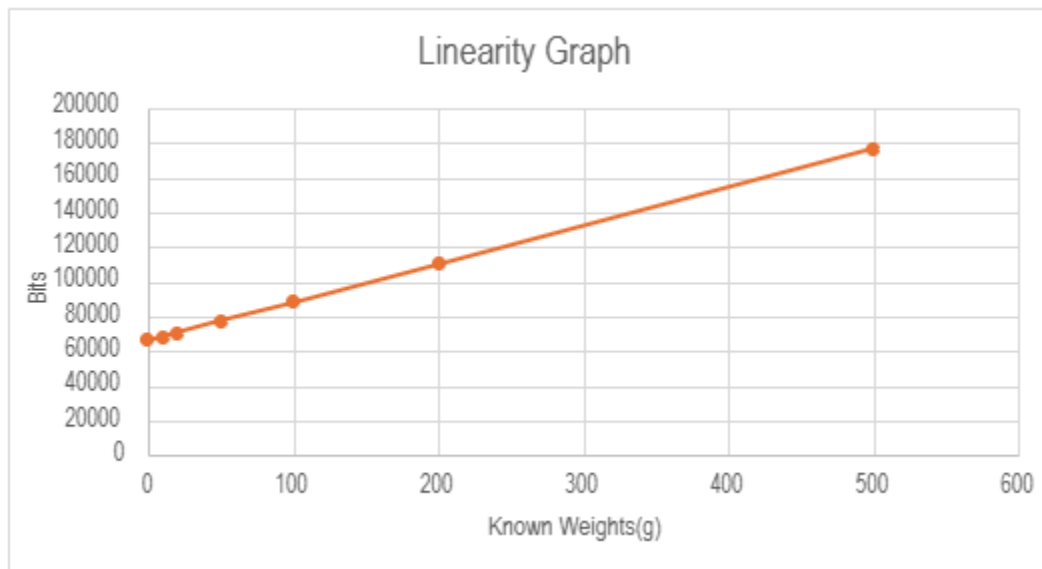
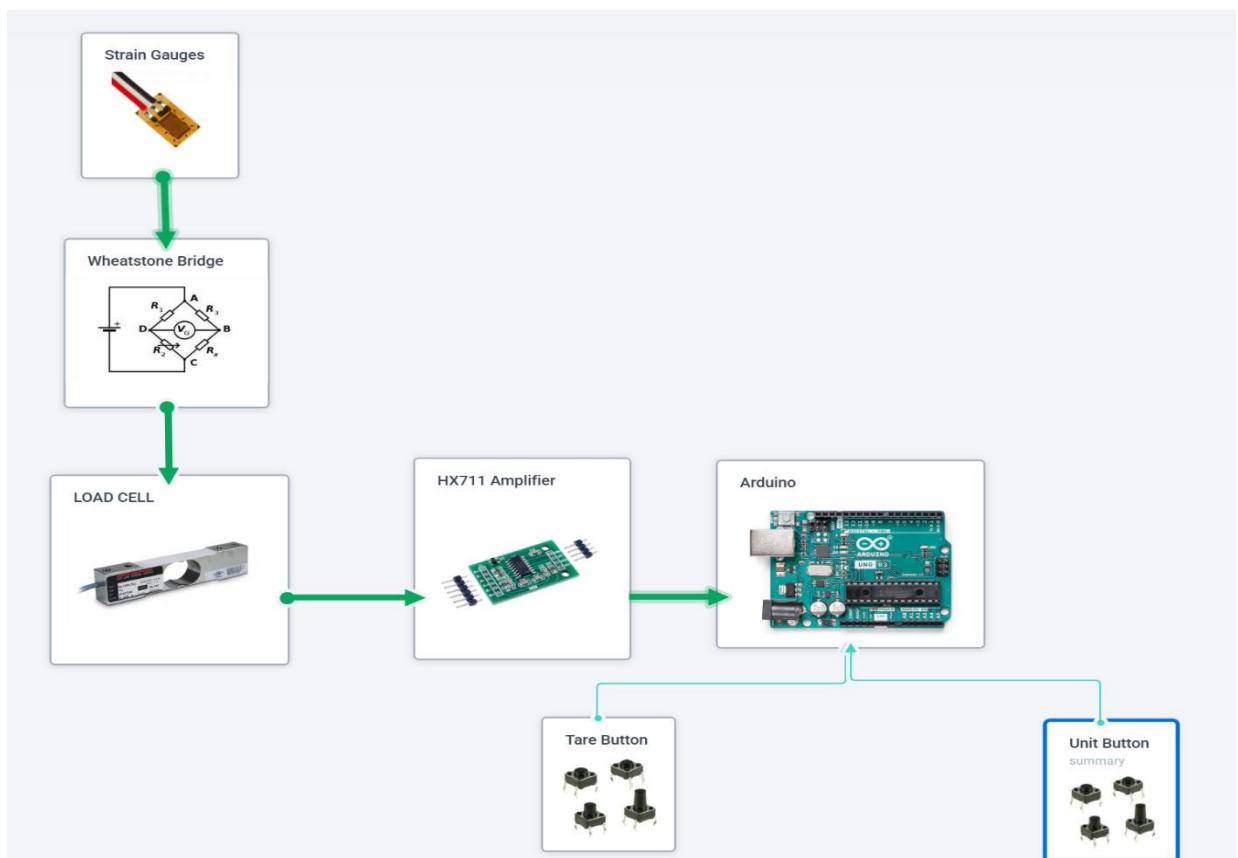


Figure 7: Linearity Graph



Known Weight	Scale Average	Percent Error	Standard Deviation	Max Load
0 g	0.45 g	0.45% g	0.24 g	9997 g
10 g	10.08 g	0.8% g	0.43 g	
20 g	10.03 g	0.13% g	0.28 g	
50 g	50.47 g	0.93% g	0.26 g	
100 g	101.07 g	1.07% g	0.21 g	
200 g	202.53 g	1.27% g	0.29 g	
500 g	506.44 g	1.29% g	0.16 g	

In order to determine max loading of the scale, the weight of the bucket(3g) was subtracted from the max load of the load cell(10kg). To test the linearity of the scale, weights of different masses were placed on the scale and their bits readings were recorded. This resulted in a linear line of the bits readings caused by different tensions on the load cell which increased as the weight increased. Three tests were done for each known weight. The weight readings the scale read were recorded and the average was recorded. Percent error calculations were done from the average weight the scale read from each known weight and standard deviation calculations were done between the three tests.

The linearity graph showing bits vs. known weights demonstrated how as weight was increased, tension in the load cell increased which increased resistance and the bits reading increased at the same rate. The linearity proved the accuracy of the scale since the scale readings increased at the same rate as the weight. The scale had a fairly low percent error of at most 1.29% difference than the actual weight. Furthermore, the standard deviation was also very low with less than one gram difference in each test. This low percent error showed that the scale was very accurate and the low standard deviation showed that it was very precise. This data including the linearity, low the scale to be accurate.

Conclusion:

Creating a scale using electrical components such as a Load Cell, Amplifier, Buttons, Capacitors, and a microcontroller was a valuable experience for the team. Team members learned to rectify mechanical issues through basic principles of electronics such as the workings of a capacitor, learned to send commands to a microcontroller to achieve accurate readings. Most importantly, the team has an understanding of Wheatstone Bridges and Amplifiers, as well as their common use cases. Through the iterative design process of physical components, electrical circuits, and programming involving a microcontroller, the Team was able to create an accurate scale and utilize the scale with real-world objects.

